

Quadratic equations and parabolas



1

- a
- i Maximum
 - ii Negative
 - iii 2 solutions
- b
- i Minimum
 - ii Positive
 - iii 0 solutions
- c
- i Maximum
 - ii Negative
 - iii 1 solution
- d
- i Minimum
 - ii Positive
 - iii 2 solutions

2 The equation has no real solution.

3

- a 2 solutions b 1 solution c 0 solutions

The discriminant and equations

4

- a
- i $\Delta = 0$
 - ii 1
- b
- i $\Delta = -76$
 - ii 0
- c
- i $\Delta = 1$
 - ii 2
- d
- i $\Delta = 16$
 - ii 2
- e
- i $\Delta = 0$
 - ii 1
- f
- i $\Delta = -23$
 - ii 0
- g
- i $\Delta = 40$
 - ii 2
- h
- i $\Delta = 100$
 - ii 2

5

- a $\Delta = 0$ b Rational

6

- a
- i $\Delta = -144$
 - ii Non-real roots.
- b
- i $\Delta = 112$
 - ii Real, unequal and irrational roots.
- c
- i $\Delta = 0$
 - ii Real, equal and rational roots.



d i $\Delta = 64$

ii Real, unequal and rational roots.

e i $\Delta = 0$

ii Real, equal and rational roots.

f i $\Delta = 65$

ii Real, unequal and irrational roots.

g i $\Delta = -4$

ii Non-real roots.

h i $\Delta = 3.75$

ii Real, unequal and irrational roots.

7 a $\Delta = 256$

b The roots are unequal and rational.

c $x = 12, x = -4$

8 There is only one real solution.

9 $b^2 - 4ac > 0$

10 a $9 + 8m$

b $16n^2 + 8$

c $45 - 4p$

d $16 + 4p^2$

e $12m$

f $\frac{-23n^2}{4}$

11 $a < \frac{9}{10}$

12 $m > -1$

13 a $m > -\frac{9}{20}$

b $m = 0$ must be eliminated as m is the coefficient of x^2 and by definition a quadratic equation must have an x^2 term.

14 $n = \pm 9$

15 $k = 1$

16 $m = -2, m = 4$

17 a

i $64 - 8k$

ii $k = 8$



iii $k \leq 8$

iv $k > 8$

v $k < 8$

b

i $400 - 8k$

ii $k = 50$

iii $k \leq 50$

iv $k > 50$

v $k < 50$

c

i $52 - 12k$

ii $k = \frac{13}{3}$

iii $k \leq \frac{13}{3}$

iv $k > \frac{13}{3}$

v $k < \frac{13}{3}$

18

a $k > 74$

b $k = 75$

The discriminant and parabolas

19 None

20 One x -intercept

21

a

i $a > 0$

ii $\Delta > 0$

b

i $a < 0$

ii $\Delta = 0$

c

i $a < 0$

ii $\Delta < 0$

d

i $a > 0$

ii $\Delta < 0$

e

i $a > 0$

ii $\Delta < 0$

f

i $a > 0$

ii $\Delta > 0$

g

i $a < 0$

ii $\Delta = 0$

h

i $a < 0$

ii $\Delta > 0$

i

i $a > 0$

ii $\Delta = 0$

j

i $a < 0$

ii $\Delta < 0$

k

i $a < 0$

ii $\Delta = 0$

l

i $a < 0$

ii $\Delta > 0$

m

i $a < 0$

ii $\Delta < 0$

n

i $a > 0$

ii $\Delta = 0$

o

i $a < 0$

ii $\Delta > 0$

p

i $a > 0$

ii $\Delta < 0$

22 a $m < -\frac{27}{4}$

b $m = -7$



23 $k = 16$

Application

24

a $0 < p < q$

b $x^2 + 2(p - q)x + p^2 - q^2 = 0$

c $\Delta = 8q(q - p)$

d Two real solutions as $q > p$, therefore the discriminant is positive.

e $x = 3p$